Choptuik's echo period at large dimension: evidence for a finite limit $\Delta_\infty = 1.64389375$, and selection beyond all orders in $1/D$

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(Dated: October 2026; unreviewed draft)

We compute the echo period $\Delta(D)$ of Choptuik's discretely self-similar critical solution of massless scalar collapse in D spacetime dimensions at 857 points $8.25 \le D \le 61.03$ with a double-double (about 32-digit) solver, consistent with smooth curves to 4×10^{-15}, and with an independent double-precision solver for $4 \le D \le 40$. One rational function of $1/(D-2)$ reproduces Δ and the critical exponent γ for all $D \ge 4$ within the precision of the points. Series in $1/D$ fitted on $D \le 45$ predict the points up to $D = 60.1$ to 5×10^{-13} (to $D = 61$: 7×10^{-13}), 40 to 300 times better than series in half-integer powers; this supports, but does not prove, analyticity in $1/D$, which then gives $\Delta_\infty = 1.64389375 \pm 10^{-8}$. A power-law approach to $\Delta = 0$, as conjectured in the literature, is excluded by the data; a logarithmic one is disfavoured. We show that the $1/D$ expansion of the self-similar field equations can be carried to all orders for any period and any time profile, so the period (and likewise γ) is selected beyond all orders in $1/D$. Without gravity the obstruction is a closed-form light-cone coefficient, which we identify as the Stokes multiplier of a mode born at large self-similar radius. As a working hypothesis, the selection rule at $D = \infty$ is the vanishing of the Stokes multipliers of an n-free inner problem; its higher-harmonic conditions predict the shape of the echo, $|F_3/F_1| = 0.055$, which the data reproduce. The harmonic-1 condition, which would fix Δ_∞, remains open. The computations, derivations and text were produced by an AI model (Claude, Anthropic) under the author's direction and have not yet been reviewed by a human expert (Sec. VII).

I. INTRODUCTION

Fine-tuning a one-parameter family of initial data for a massless scalar field to the threshold of black-hole formation, Choptuik found that the evolution approaches a universal critical solution that repeats itself on ever smaller scales: it is discretely self-similar (DSS) with echo period Δ, and the masses of black holes formed just above threshold scale as $M \propto (p - p^*)^\gamma$ [1]. In four dimensions $\Delta = 3.445452402(3)$ [3] and $\gamma \simeq 0.374$ [2]; see the review [4].

The dependence on the spacetime dimension D has been studied numerically, mostly at percent-level precision, for integer $D = 4$–14 [5–9], and to about 10^{-6} for continuous $3.05 < D < 5.5$ by Ecker, Ecker, Grumiller and Jechtl [10], who find a maximum of Δ near $D = 3.756$. Ecker, Ecker and Grumiller [11] constructed analytic DSS solutions at large D and, with [10], conjectured $\Delta \to 0$ and $\gamma \to 1/2$ as $D \to \infty$. Large D has been a fruitful limit in gravity [12].

This paper reports the following.

1. $\Delta(D)$ at 857 points $8.25 \le D \le 61.03$ from a double-double solver, and Δ, γ for $4 \le D \le 40$ from a double-precision solver; one rational formula in $1/(D-2)$ for all $D \ge 4$ (Sec. III). It reproduces the published values at $D = 4$ and the maximum of Δ below $D = 4$; at $D = 5$ there is an unresolved 6×10^{-5} discrepancy with [10].

2. Held-out tests that support analyticity of Δ in $1/D$, giving $\Delta_\infty = 1.64389375 \pm 10^{-8}$; a power-law approach to zero is excluded and a logarithmic one disfavoured; nine candidate closed forms for Δ_∞ are ruled out (Sec. IV).

3. The $1/D$ expansion never fixes the period (Proposition 1); the leading-order problem and its light-cone coefficient in closed form (Proposition 2); what gravity does to the non-analytic piece (Proposition 3); and a derivation identifying that coefficient as a Stokes multiplier (Proposition 4) (Sec. V).

4. An n-free inner problem whose Stokes multipliers, by a working hypothesis, select the period; its higher-harmonic conditions predict $|F_3/F_1| = 0.055$ and $|F_5/F_1| = 0.0030$ for the echo profile at $D = \infty$, against 0.0549–0.0552 and 0.0032–0.0034 extrapolated from the solutions (Sec. V F). The harmonic-1 condition, which would give Δ_∞, is open.

5. An analysis of the next-to-leading-order condition of [11], which within this framework does not select Δ (Sec. V G).

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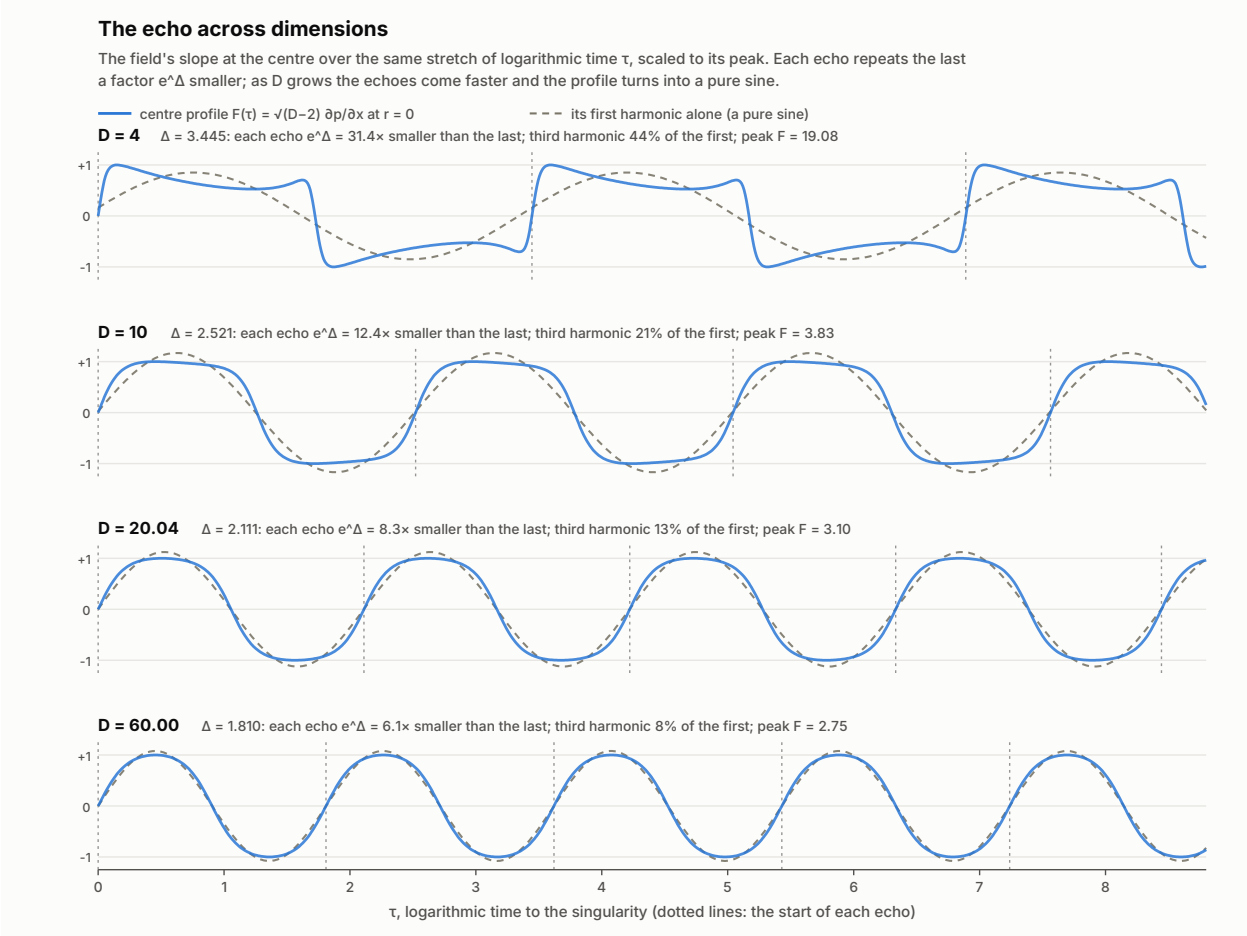


FIG. 1. The echo across dimensions: the centre slope of the field, $F(\tau) = \sqrt{D-2}\,\partial_x p$ at $r = 0$, over the same stretch of logarithmic time τ for $D = 4$, 10, 20 and 60, each scaled to its peak (dashed: its first harmonic). At $D = 4$ the profile is close to a square wave and each echo is $e^{\Delta} = 31\times$ smaller than the last; at $D = 60$ it is nearly a pure sine and the factor is 6.1. The $D = 4$ profile is from the double-precision solver (39 harmonics); the others are double-double solutions.

II. EQUATIONS AND NUMERICAL METHOD

We use the self-similar radius x (radius over time to the singularity, normalized so that the past light cone of the singularity is at $x = 1$) and $\tau = -\ln(-t)$, or the phase $\theta = \omega\tau$ with $\omega = 2\pi/\Delta$. With $n = D - 2$, p and m the (scaled) derivatives of the scalar field along the two radial null directions, a the radial metric function ($b = a^2$) and c the lapse-type function that sets the light cones, the equations are

$$x c_x = (n-1)\,c\,(b-1), \qquad x b_x = b\,(p^2+m^2) - (n-1)\,b\,(b-1), \tag{1}$$

$$x(x-c)\,p_x = -\omega x\,p_\theta + Gp - cp + \tfrac{n}{2}\,c\,(p+m), \qquad x(x+c)\,m_x = -\omega x\,m_\theta - Gm + cm - \tfrac{n}{2}\,c\,(p+m), \tag{2}$$

with $G = (n-1)\,c\,(b-1)$, on $0 \le x \le 1$; p, m contain only odd harmonics of θ and b, c only even ones; the solution is regular at $x = 0$, analytic at the past light cone $x = 1$, and $c(1) = 1$ fixes the gauge.

Double-precision solver ($4 \le D \le 40$): spectral multiple shooting in x with Fourier modes in τ (23–47 modes), with the integration-step error extrapolated out; near $D = 4$ the points are extrapolated in the mode count. Above $D \approx 26$ the period is selected by an effect that weakens exponentially with D, and rounding at 10^{-16} moves Δ by 10^{-7}–10^{-6} at $D = 30$–40.

Double-double solver ($8.25 \le D \le 61.03$): double-double arithmetic (about 32 digits) on SIMD vectors; Taylor series in x of order 36 with truncation 10^{-32}; centre and horizon series to 10^{-33}; collocation in τ on the half period (32 points); an exact Jacobian by forward-mode duals with a double-double LU; continuation in D with deflated Newton steps. The Jacobian has a ladder of near-null directions (high τ-harmonics of the centre data, singular values falling roughly like $e^{-0.7k}$); from $D = 35$ the centre harmonics above 13 are pinned, which moves Δ by 5×10^{-15} at

TABLE I. Checks of the double-double solver.

Check	Result
double-double arithmetic against mpmath	1.3–1.6×10^{-32} relative
horizon series against 40-digit mpmath	2×10^{-32}
Newton at $D = 8$	quadratic, $\lvert F\rvert = 5 \times 10^{-31}$ in 4 steps
$\Delta(7)$, $\Delta(7.75)$ against 39-mode references ($\pm 3 \times 10^{-11}$)	$+3.1 \times 10^{-12}$, -2.8×10^{-12}
32 / 48 / 64 collocation points in τ, $D = 7$–17	Δ agrees to 10^{-16}
32 / 48 collocation points, $D = 40$	9×10^{-22}
harmonics pinned above 13 / above 15, $D = 40$	3×10^{-17}
second machine (separate build), $D = 59.84$–60.05	2–4×10^{-16}
32 points $D = 60.95$–61.03 against the $1/D$ fit on $D \le 60.1$	2.6×10^{-15}

TABLE II. Coefficients of Eq. (3).

k	a_k	b_k	e_k	f_k
0	$+1.6438975180$	1	$+0.5$ (fixed)	1
1	$+4.0378045077 \times 10^{1}$	$+1.8251453626 \times 10^{1}$	$+1.8083629635 \times 10^{1}$	$+3.7683686025 \times 10^{1}$
2	$+2.5233285989 \times 10^{2}$	$+6.8182050167 \times 10^{1}$	$+1.5241588758 \times 10^{2}$	$+3.4245505235 \times 10^{2}$
3	$+2.0673502321 \times 10^{2}$		$-5.2341588541 \times 10^{1}$	
4	$-3.5709850575 \times 10^{2}$		$+9.9259594505$	
5	$+3.1347320996 \times 10^{2}$		$-9.8386450353 \times 10^{1}$	
6	$-4.1891374574 \times 10^{2}$		$+7.4529506074 \times 10^{1}$	
7	$+3.2695252782 \times 10^{2}$		$-4.8272358903 \times 10^{1}$	
8	$-1.6050467699 \times 10^{2}$			

$D = 35$ and 3×10^{-17} at $D = 40$. Checks are listed in Table I. The 857 points fit smooth curves to about 4×10^{-15}; the estimated error per point is about 10^{-16} to $D = 60.1$ and a few 10^{-15} above, where the tolerance was relaxed.

III. RESULTS

A. One formula for all $D \ge 4$

With $\varepsilon = 1/n = 1/(D-2)$,

$$\Delta(D) = \frac{\sum_{k=0}^{8} a_k \varepsilon^k}{1 + b_1\varepsilon + b_2\varepsilon^2}, \qquad \gamma(D) = \frac{\frac{1}{2} + \sum_{k=1}^{7} e_k \varepsilon^k}{1 + f_1\varepsilon + f_2\varepsilon^2}, \tag{3}$$

with the coefficients of Table II, fitted on the double-precision points $4 \le D \le 30.2$. They reproduce Δ to 5×10^{-10} ($4 \le D \le 4.3$), 4×10^{-9} (4.3–20) and 10^{-8} (20–30), and γ to 3.4×10^{-7}, 3.3×10^{-8} and 2.3×10^{-9}, each within the precision of the points. Against the double-double points, which the fit never saw, the formula holds to 2.7×10^{-9} on 8.25–21, 7.5×10^{-9} to $D = 30$ and 5.8×10^{-8} to $D = 61$ (Fig. 2). The limit $\gamma_\infty = 1/2$ is imposed in the γ fit, not derived; unconstrained fits on $D \ge 8$–12 give 0.4996–0.4998. The denominators vanish only at $D = -3.2, -11.0$ (Δ) and $D = -13.3, -20.4$ (γ). Values are listed in Table III.

B. Comparison with published values

At $D = 4$ the formula gives $\Delta = 3.4454524$ and $\gamma = 0.3739610$, in agreement with $\Delta = 3.445452402(3)$ [3] to about 10^{-8} and with $\gamma = 0.373961 \pm 10^{-6}$ [10] to about 10^{-7}. Below $D = 4$ a cubic through the 47-mode double-precision points at $D = 3.70$–3.80 puts the maximum of Δ at $D = 3.75570$, $\Delta = 3.466774$, in agreement to 2×10^{-6} with $D = 3.755726$, $\Delta = 3.466772$ of [10].

At $D = 5$, Ref. [10] gives $\Delta = 3.22176$ and $\gamma = 0.41322$, which are 5.8×10^{-5} and 9.3×10^{-5} above our 3.2217019 and 0.4131269. Ours are stable across 31, 39 and 47 Fourier modes to about 10^{-7}, and Bland and Kunstatter give $\gamma(5) = 0.4131 \pm 0.0001$ [8], matching ours. The cause of the discrepancy is not known; an independent recomputation is in progress. Below $D = 4$, γ extrapolated from our 47-mode points suggests $\gamma(3.5) \lesssim 0.33$, against 0.349 ± 0.003 in [7]; our points there are under-resolved, and this is an open check.

TABLE III. Values. Δ for $D = 10$–60 is double-double (interpolated to 10^{-14} where D is not a computed point); $\Delta(8)$ is a separate double-double run; Δ for $D = 4$–7 and γ to $D = 40$ are double-precision; $\gamma(50)$, $\gamma(100)$, $\gamma(1000)$ are formula values. $\Delta(70)$ and $\Delta(100)$ are extrapolations on which all models fitting the double-double points agree (to 10^{-11}, 2×10^{-8}); $D = 1000$ and ∞ assume analyticity in $1/D$.

D	Δ	γ	D	Δ	γ
4	3.4454524	0.3739610	20	2.112080154580	0.4737107
5	3.2217019	0.4131269	30	1.964957899258	0.4807927
6	3.0094832	0.4300795	40	1.888568197910	0.484776
7	2.8427532	0.4399912	50	1.841659503007	0.48736
8	2.711790054444	0.4467164	60	1.809889459975	
10	2.521078919562	0.4555879	70	1.78693476494	
12	2.389188553641	0.4614056	100	1.7450264844	0.49313
15	2.253098407194	0.4673656	1000	1.65424086	0.4992
			∞	1.64389375 ± 10^{-8}	1/2 (assumed)

TABLE IV. Held-out test of large-D models (cleaned double-double points; snapshot $D \le 60.1$).

Model	max residual, $10 \le D \le 45$	worst error, $45 < D \le 60.1$	fitted on $D \le 37.5$: worst error	limit
$1/D$, degree 16	4.1×10^{-15}	5.2×10^{-13}	4.2×10^{-11}	1.64389375
$1/D$, degree 18	4.0×10^{-15}	1.8×10^{-12}	1.5×10^{-10}	1.64389377
$1/(D-2)$, degree 18	4.0×10^{-15}	2.8×10^{-12}	7.9×10^{-11}	1.64389377
$(D-2)^{-1/2}$, degree 20	3.9×10^{-15}	2.0×10^{-11}	3.9×10^{-10}	1.636
$D^{-1/2}$, degree 20	3.8×10^{-15}	1.6×10^{-10}	6.2×10^{-9}	1.547

C. Physical reading

Each echo is a factor e^{Δ} smaller than the last: 31 at $D = 4$, 12 at $D = 10$, 5.2 at $D = \infty$. A near-critical collapse makes about $\gamma\ln(1/|p-p^*|)/\Delta$ echoes, so each further echo costs $\Delta/(\gamma\ln 10)$ decimal digits of fine-tuning: 4.0 at $D = 4$, 2.4 at $D = 10$ and 1.43 at $D = \infty$. At large D the field and its gravity weaken: the centre slope of the field falls like $2.72/\sqrt{n}$ and the departure of c from 1 at the centre like $3.65/n$ (Sec. V).

IV. THE LARGE-D LIMIT

A. Is the period analytic in $1/D$?

Each model in Table IV is a polynomial in one variable, fitted at 60 digits on the cleaned double-double points $10 \le D \le 45$ (or ≤ 37.5) and compared with every point above, up to $D = 60.1$, which it never saw (the data snapshot when the table was made; Fig. 3 uses all points to $D = 61.0$). Every model fits its range to the level of the points; the unseen points decide. Fitted on $D \le 45$, the $1/D$ series predicts Δ fifteen units of D ahead to 5×10^{-13}, while the series in half-integer powers miss by 40 to 300 times more, and their limits jump with the degree. Adding one non-analytic term ($D^{-1/2}$, $D^{-3/2}$, $\ln D/D$, $\ln D/D^2$ or $e^{-D/4}$) to the degree-16 series changes neither the prediction nor the limit, and the term's coefficient is not determined.

Fitted on all cleaned points, power series in $1/D$ of degree 16–20 and in $1/(D-2)$ of degree 18–20 fit to the noise and give limits from 1.643893744 to 1.643893752 (snapshot $D \le 60.1$) and from 1.643893745 to 1.643893752 (all 823 cleaned points to $D = 61.03$). Hence, if Δ is analytic in $1/D$,

$$\Delta_\infty = 1.64389375 \pm 10^{-8}, \qquad \omega_\infty = 2\pi/\Delta_\infty = 3.82213589 \pm 2\times10^{-8}, \tag{4}$$

and $\Delta(D) \approx \Delta_\infty + 10.37/n + \dots$. The analytic form is the natural default but not proven: as shown in Sec. V, the period is selected by an effect the $1/D$ expansion does not see.

B. The conjecture $\Delta \to 0$

The conjecture $\Delta \to 0$ [10, 11] comes with no rate. Fitted on $10 \le D \le 45$ and compared with $45 < D \le 61$: a power-law approach to 0, D^{-p} times a series in $1/D$ with $p \ge 0.01$, misses the held-out points by 6×10^{-8} or more,

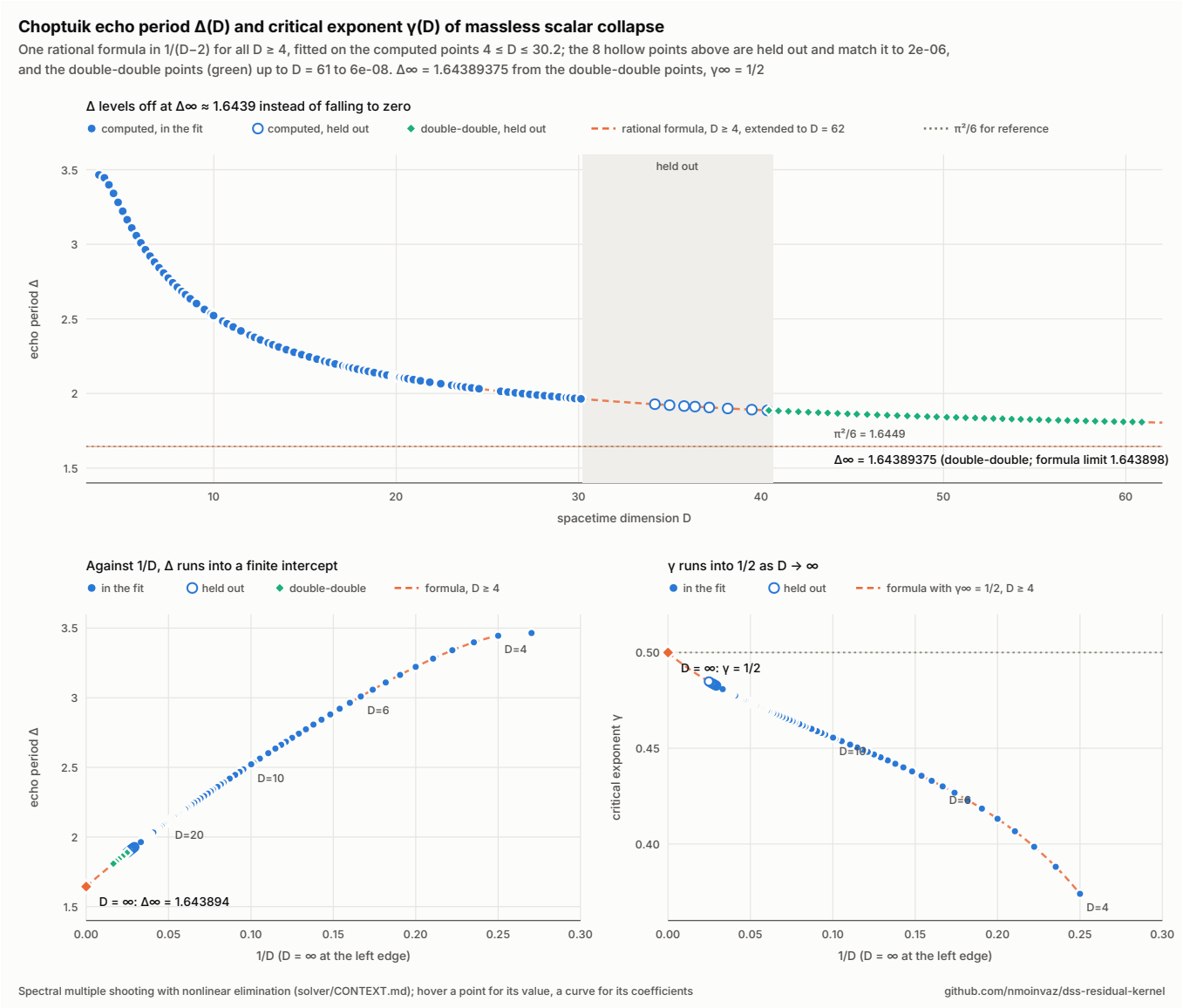


FIG. 2. $\Delta(D)$ and $\gamma(D)$ with the rational formula (3) (dashed), fitted on $4 \le D \le 30.2$; hollow points are held out, and green diamonds are double-double points up to $D = 61$.

against 7×10^{-13} for the analytic series; a series in $1/\ln D$ with zero limit misses them by 1.3×10^{-11} and still gives $\Delta \approx 0.88$ at $D = 10^{30}$; adding a $c/\ln D$ term to the analytic series gives $c \approx 4 \times 10^{-7}$ and leaves the limit at 1.6438937. A power-law approach to 0 is excluded; a logarithmic one is disfavoured but cannot be excluded from $D \le 61$ alone. Bland's hypothesis $\Delta_\infty = 5/2$ [9] is excluded by $\Delta(14) = 2.29$ and decreasing.

C. Candidate closed forms

Nine closed forms were forced as the limit of a power series in $1/D$ fitted on $10 \le D \le 45$ (degree 14 to 26) and compared with the 351 cleaned points from 45 to 59.8 (Table V). The closest sits 1.3×10^{-5} from the limit, more than 1000 times its uncertainty, and every forced fit predicts the unseen points at least 100 times worse than the free fit. A search over products of standard constants and Bessel zeros finds nothing in 1.6438936–1.6438939, nor in a window about 27 times wider. Whether Δ_∞ has a closed form is open.

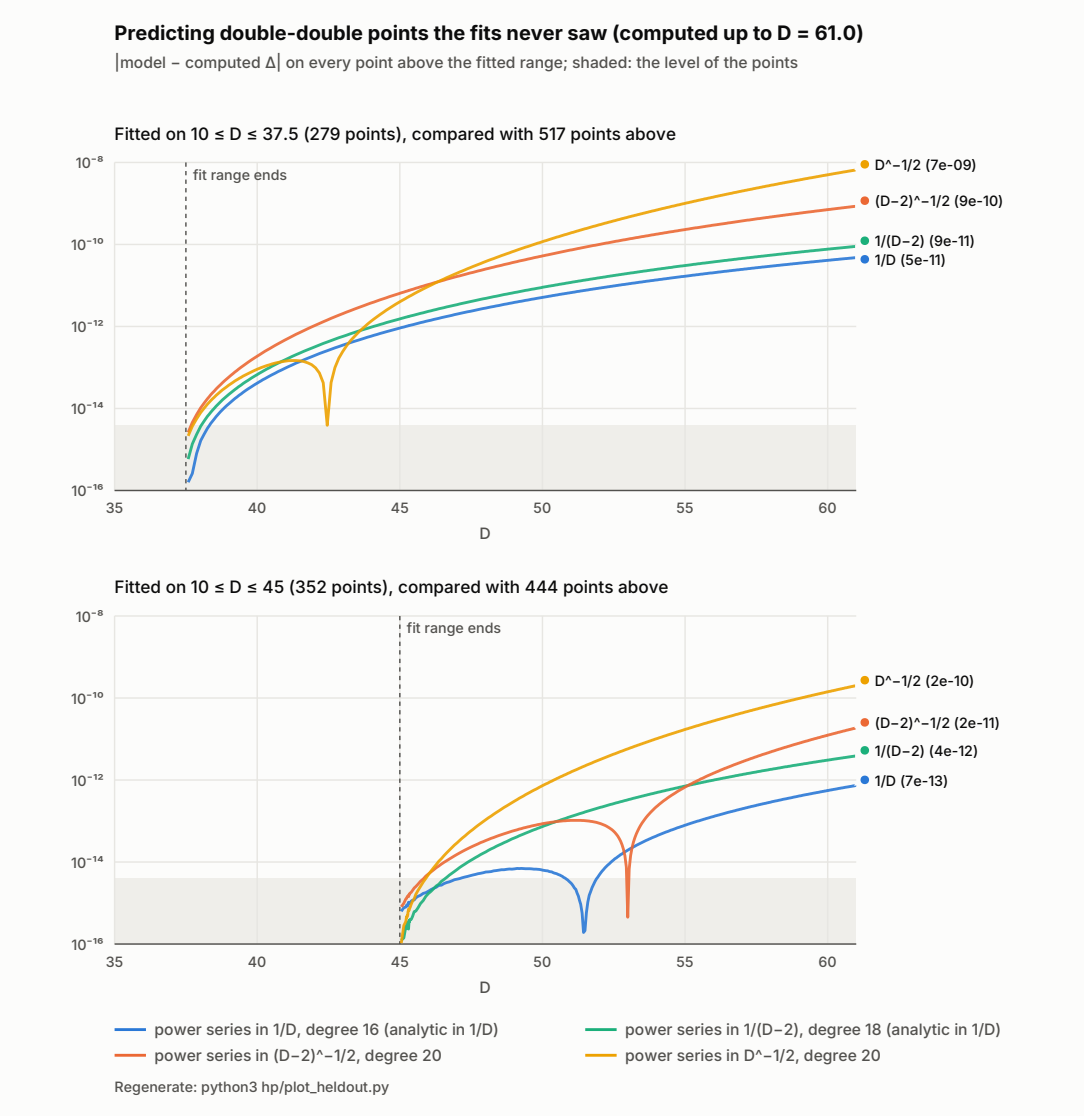


FIG. 3. |model − computed Δ| on every double-double point above the fitted range, for fits on $10 \le D \le 37.5$ (top) and $10 \le D \le 45$ (bottom); shaded: the level of the points.

V. LARGE-D THEORY

A. Scaling

On the double-double solutions at $D = 12$–60: $p \approx xF(\theta)/\sqrt{n}$ near the centre and $p + m$ is smaller by a further factor of order $1/n$; $c = 1 - O(1/n)$ and $b - 1 = (p^2 + m^2)/(n - 1)$ to about 3%; and the centre series has the exact symmetry $m(x) = p(-x)$. The first harmonic of $F = \sqrt{n}\,\partial_x p(0)$ falls from 3.480 at $D = 20$ to 2.962 at $D = 60$ and extrapolates to about 2.72; $n(1 - \langle c(0)\rangle)$ and $\langle F^2\rangle$ approach a common limit near 3.7.

B. The $1/n$ expansion never fixes the period

Write $p = \varepsilon^{1/2}P$, $m = \varepsilon^{1/2}M$, $S = P + M$, $\Delta_P = P - M$, and expand $S = \sum_{k\ge1}\varepsilon^k S_k$, $\Delta_P = \sum_{k\ge0}\varepsilon^k D_k$, $c = 1 - \sum_{k\ge1}\varepsilon^k G_k$, $b = 1 + \sum_{k\ge2}\varepsilon^k B_k$, $\omega = \sum_k \varepsilon^k \omega_k$ (integer powers of ε; this is the scaling measured above, and the statements below hold within this ansatz).

TABLE V. Candidate closed forms for Δ_∞, forced as the limit: best held-out error and its ratio to the free fit. G is Catalan's constant, $j_{\nu,k}$ the k-th zero of J_ν.

Candidate	value	minus Δ_∞	held-out error (ratio)
none (free fit)	1.64389375 ± 10^{-8}		4.7×10^{-13}
$\frac{5}{7}\sqrt{G}\, j_{0,1}$	1.6439747	$+8.1 \times 10^{-5}$	6.2×10^{-11} (130)
$\frac{7}{5}(\ln 2)^{-2}\pi^{-1/2}$	1.6440014	$+1.1 \times 10^{-4}$	1.1×10^{-10} (220)
$\frac{7}{2}\sqrt{\ln 2/\pi}$	1.6440152	$+1.2 \times 10^{-4}$	1.3×10^{-10} (270)
$\frac{9}{8}\sqrt{j_{2,1}/j_{0,1}}$	1.6440207	$+1.3 \times 10^{-4}$	1.4×10^{-10} (290)
$\frac{10}{9}\sqrt{j_{0,1}/\ln 3}$	1.6439063	$+1.3 \times 10^{-5}$	5.4×10^{-11} (110)
$\pi^2/6$	1.6449341	$+1.0 \times 10^{-3}$	1.7×10^{-9} (3500)
$2\pi/j_{1,1}$	1.6397878	-4.1×10^{-3}	7.0×10^{-9} (15000)
$\sqrt{e}$	1.6487213	$+4.8 \times 10^{-3}$	8.1×10^{-9} (17000)
$(1+\sqrt{5})/2$	1.6180340	-2.6×10^{-2}	4.4×10^{-8} (92000)

Proposition 1 *Let $F(\theta)$ be any smooth 2π-periodic function with odd harmonics only, $\omega_0, \omega_1, \ldots$ any numbers and $K_1(\theta), K_2(\theta), \ldots$ any smooth odd-harmonic functions. Then there is exactly one formal solution with $D_0 = 2xF$ in which, at every order, S_k is an even polynomial in x starting at x^2, $D_k - xK_k$ is an odd polynomial starting at x^3, and G_k, B_{k+1} are even polynomials (B starting at x^2) with $G_k(1) = 0$. Every order is regular at the centre and analytic at the light cone.*

Proof sketch. Induction on k. The b equation gives B_{k+1} algebraically (its coefficient is $(n-1)b \approx 1/\varepsilon$); the c equation gives $-x\partial_x G_k \in x^2 E$ (E: even polynomials with periodic coefficients), so G_k is an even polynomial without $\log x$, fixed by $G_k(1) = 0$; the difference equation gives $S_k \in x^2 E$; and the sum equation reads $x\partial_x D_k - D_k = R_k$ with R_k odd and starting at x^3, because $x\partial_x - 1$ annihilates xK_j. The only possible obstruction, an x^1 term in R_k that would produce $x \log x$, never occurs. Parity in x and the harmonic content in θ are preserved at each step. Polynomials are analytic at $x = 1$, and nothing constrains F or the ω_k. □

The first orders (sympy, to fourth order) are $P_0 = xF$, $B_2 = 2x^2F^2$, $G_1 = F^2(1-x^2)$, $S_1 = 2x^2(F + \omega_0 F')$ and $D_1 = x^3(\omega_0^2 F'' + 3\omega_0 F' + 2F - 2F^3) + xK_1$. Substituting the expansion through S_3, G_3, B_4 into the exact equations with F from the $D = 60$ solution, the relative residuals fall like n^{-3} for $n = 50$–800, both for $\omega_0 = 2\pi/1.64389$ and for a wrong $\omega_0 = 2\pi/1.4$.

So if $\Delta(D)$ is analytic in $1/D$, its expansion coefficients cannot be computed order by order from the field equations: the selection works through terms smaller than any power of $1/n$. The same holds for the critical exponent: the expansion of growing perturbations $e^{\lambda\tau}$ goes through for any λ (the same induction with $\partial_\tau \to \lambda + \omega\partial_\theta$; checked with sympy through order 2), so $\gamma = 1/\lambda$ is also selected beyond all orders.

C. The leading-order problem

At leading order ($c = b = 1$) the equations describe a free wave in self-similar form. For one harmonic $e^{i\mu\theta}$ put $s = i\mu\omega$, $\alpha = (1+s)/2$, $\beta = (2+s)/2$, $\kappa = (n+1)/2$.

Proposition 2 *The solution regular at the centre is, up to a constant, $p - m = x\,{}_2F_1(\alpha, \beta; \kappa; x^2)$. Near the light cone it contains the non-analytic branch $(1-x^2)^{n/2-1-s}$ with coefficient*

$$C(n,s) = \frac{\Gamma\left(\frac{n+1}{2}\right)\Gamma\left(s+1-\frac{n}{2}\right)}{\Gamma\left(\frac{1+s}{2}\right)\Gamma\left(\frac{2+s}{2}\right)}, \tag{5}$$

which is finite and non-zero for every real $\omega \neq 0$, odd μ and $n \geq 2$. Hence no time-periodic self-similar solution of the linear problem is analytic at the light cone, for any $D \geq 4$.

Proof sketch. Eliminating $p + m$ gives Gauss's equation for $y = x\,w(x^2)$; regularity (for $n = 2$, parity) selects ${}_2F_1(\alpha, \beta; \kappa; z)$, and the connection formula [14] applies because $\kappa - \alpha - \beta = n/2 - 1 - s$ is not an integer; its second coefficient is $C(n,s)$, a ratio of Gamma functions whose arguments have non-zero imaginary parts. □

For the fundamental at $\omega = \omega_\infty$, $|C| = 0.0121, 0.0114, 0.0124, 0.0192, 0.0405$ at $n = 18, 38, 58, 200, 1000$. The free wave equation in similarity coordinates was analysed in 3+1 dimensions by Donninger [13]; the leading-order freedom of the large-D problem is also noted in [11].

D. What gravity does to the non-analytic piece

The non-analytic piece is a fast WKB mode; in Proposition 2 it is $x^{2-n}(1-x^2)^{n/2-1-s}$.

Proposition 3 *Let $c = 1 - \varepsilon F^2(1-x^2) + O(\varepsilon^2)$ as in Proposition 1, and $\Phi(x) = \ln x - \frac{1}{2}\ln(1-x^2)$. Perturbations $\delta p = e^{-n\Phi}(A + O(1/n))$, $\delta m = e^{-n\Phi}\big(\frac{1-x}{1+x}A + O(1/n)\big)$ solve the linearized equations on $0 < x < 1$ to this order if and only if*

$$x(1-x^2)\,\partial_x A - 2\omega x^2\,\partial_\theta A = (1 + x - 2x^2F^2)\,A, \tag{6}$$

i.e. along the characteristics $\theta = \theta_0 + \omega\ln(1-x^2)$, $A = A_0(\theta_0)\,\frac{x}{1-x}\exp\big(-\int_0^x \frac{2tF(\theta_0+\omega\ln(1-t^2))^2}{1-t^2}\,dt\big)$.

Proof sketch. The order-n terms give $x(1-x^2)\Phi' = 1$ and $\delta m/\delta p = (1-x)/(1+x)$; at order 1 the corrections satisfy a singular linear system whose left null vector $(1+x, -(1-x))$ gives the transport equation as the solvability condition. Perturbations of b, c and $G\,\delta p$ enter one order later (confirmed symbolically from the exact equations). □

Integrating the linearized equations directly from $x = 0.9$ to 0.72 ($F = 2.72\sin\theta + 0.22\sin 3\theta$, $\omega = \omega_\infty$) reproduces the transported A with relative errors 0.57, 0.26, 0.13, 0.062, 0.031 at $n = 100$–1600, i.e. like $1/n$; with the full $O(1/n)$ background the numbers are nearly the same. Three consequences: (i) gravity acts on the non-analytic piece at order 1 (over this interval it multiplies it by about 40), and shifts the light-cone exponent by $\langle F^2\rangle \approx 3.7$, so an expansion in the wave amplitude is not a route to the selection; (ii) the factor never vanishes, so gravity cannot cancel the piece on the way to the light cone; (iii) every characteristic reaches $x = 0$, where the fast mode is the singular branch x^{2-n}, so a regular solution carries no fast mode on $0 < x < 1$ at any order of this WKB expansion. The non-analytic piece must come from a Stokes phenomenon.

E. The light-cone coefficient as a Stokes multiplier

Proposition 4 (derivation) *Let $y(\varepsilon) = x\,{}_2F_1(\alpha,\beta;\kappa;z)$ with $z = x^2$, $\kappa = (1+1/\varepsilon)/2$, and $y \sim \sum_k y_k\varepsilon^k$. Then*

$$y_k = 2x(-1)^{k-1}\sum_{J\ge 0} R_J(2J+1)^{k-1}, \tag{7}$$

$$R_J = \frac{(-1)^J(\alpha)_{J+1}(\beta)_{J+1}}{J!\,(J+1)!}\,z^{J+1}\,{}_2F_1(\alpha+J+1,\beta+J+1;J+2;z), \tag{8}$$

convergent for $x < 1/\sqrt{2}$ and continued above, and as $k\to\infty$ $y_k \approx \sum_\ell K_\ell(x)\,\Gamma(k+s+\frac{1}{2})/\big(\Phi(x)+i\pi(\ell+\frac{1}{2})\big)^{k+s+1/2}$, dominated by $\ell = -1$, i.e. by the singulant $\chi = \Phi - i\pi/2$.

The first formula is the pole expansion of ${}_2F_1$ in κ; the second follows from the large-J form of R_J and Poisson summation over J. This is a derivation, not a proof: a full proof needs a growth bound for the pole expansion, the exchange of the J-sum with the expansion of $1/(1+(2J+1)\varepsilon)$, error bounds inside the Poisson sum, control of the continuation to $x > 1/\sqrt{2}$, and the branch of $\chi^{k+s+1/2}$. Numerically, the pole sum equals the direct series to 10^{-49}–10^{-91}; the late orders computed to order 122 at 90 digits give $\chi = \Phi - i\pi/2$ to 3–7 $\times 10^{-3}$ at $x = 0.7$–0.9; and the predicted Dingle constant $K = e^{-i\pi(1+s)}\,2^{-s-1/2}/(\Gamma(\alpha)\Gamma(\beta))$, with prefactor $x^2(1-x^2)^{-1-s}$ and exponent $g = -(s+\frac{1}{2})$ [15], is matched to 6×10^{-4} at $x = 0.9$.

Hence: $e^{-n\chi} = e^{in\pi/2}x^{-n}(1-x^2)^{n/2}$ is the fast mode of Proposition 3; χ vanishes only at $x = \infty$, so the mode is born at the outer singular point; its Stokes line is the real axis $x > 1$ approached from above, where $\chi = \ln(x/\sqrt{x^2-1})$; on $0 < x < 1$ nothing is switched on (the optimally truncated series matches the exact solution to 4×10^{-40} at $x = 0.9$, $n = 60$, against a singular-branch size of 7×10^{-21}). The Stokes line ends at the light cone, and $C(n,s)$ is the Stokes multiplier.

F. The inner problem

At $x \sim \sqrt{n}$ the expansion is not uniform: gravity is of order one and fast and slow modes are not separated. With $\xi = x/\sqrt{n}$, $y = p - m$, $\sigma = \sqrt{n}\,(p+m)$, $b - 1 = \eta/n$ and $L = \xi^2\partial_\xi + \omega\xi\partial_\theta$, the equations become, at leading order,

$$L\,y = c\,\sigma, \qquad L\,\sigma = c\,\xi\,\partial_\xi y + c\,(\tfrac{1}{2}y^2 - 1)\,y, \qquad \xi\,\partial_\xi c = \tfrac{1}{2}\,c\,y^2, \qquad \eta = \tfrac{1}{2}y^2, \tag{9}$$

with no large parameter left. Its formal solution regular at $\xi = 0$ reproduces the top-degree coefficients of the outer polynomials of Proposition 1 (checked to 2×10^{-13} at orders 1–10). Without gravity, with $t = 1/\xi$ and $w = t^{-s}y$ along characteristics, it reduces to Hermite's (Weber's) equation; the regular solution is $y = 2\xi\,{}_2F_0(\alpha, \beta; ; 2\xi^2)$, whose Stokes constant gives, by the duplication formula,

$$C(n,s) \approx -\,i\,\frac{\sqrt{2\pi}}{\Gamma(1+s)}\,(n+1)^{s+1/2}\,e^{-i\pi\Lambda}\,\big(1+O(s^2/n)\big), \qquad \Lambda = \tfrac{n}{2} - 1 - s, \tag{10}$$

Weber's classical Stokes constant [16]; the ratio to the exact (5) is 1.00014 at $n = 10^5$ ($\mu = 1$).

Working hypothesis. With gravity, the selection rule at $D = \infty$ is that the fast-type Stokes multipliers of the formal inner solution vanish, one complex condition per odd harmonic; the unknowns are F's harmonics and ω, so the count balances. This is derived from the matching but not proved. In the inner region the fast mode obeys $\xi\partial_\xi a + 2\omega\partial_\theta a = 2F^2 a$ (the large-x limit of Proposition 3); the Borel singularity stays at $\zeta = \frac{1}{2}$ and the exponents shift by $\langle F^2\rangle$.

Computations with gravity. Two independent implementations (lateral solutions on either side of the Stokes line, and Borel–Darboux fits in ball arithmetic) reproduce the linear constant to 10^{-11} and 6×10^{-27}. With $F = A\sin\theta$ alone, both find an isolated zero of the multiplier at $A = 2.542148$, $\omega = 2.969777$; it is not the physical solution (ω 22% low) and does not survive adding harmonic 3. Truncating harmonics is not a controlled approximation here: the linear multipliers grow like $F_\nu e^{\pi|\nu|\omega/2}$.

Amplitude expansion. Writing $F = A\sin\theta + \sum_{\nu\geq 3} f_\nu(A)e^{i\nu\theta}$ (plus c.c.), harmonic μ first appears at order $A^{|\mu|}$, so every harmonic is kept exactly. Tuning f_ν order by order so that the multipliers of all harmonics $|\nu| \geq 3$ vanish, at $A = 2.72$ and $\omega = \omega_\infty$, gives $|F_3/F_1| = 0.055$ and $|F_5/F_1| = 0.0030$, with ratios $f_{\nu+2}/f_\nu \approx qA^2$, $|q|A^2 \approx 0.063$, nearly independent of ν. The double-double solutions at $D = 20$–60, extrapolated in $1/n$ (degree 2–3), give $|F_3/F_1| \to 0.0552$–0.0549 and $|F_5/F_1| \to 0.0034$–0.0032, and the ratios $|F_{\nu+2}/F_\nu|$ ($\nu = 1, \ldots, 9$) extrapolate to 0.055–0.072, a finite decay rate of 1.3–1.45 per harmonic, as predicted (Fig. 4; the extrapolations are not fully converged). This is the first number the inner problem has produced that can be checked against the solutions, and it agrees. The harmonic-1 condition, which would fix ω and hence Δ_∞, is still open: its series $S_1/S_1^{(1)} = 1 + \sigma_1 A^2 + \sigma_2 A^4 + \ldots$ is known to four terms ($\sigma_1 = 0.7436 + 0.5245i$, $\sigma_2 = 0.1188 + 0.3981i$, $\sigma_3 = -0.0540 + 0.1166i$, $\sigma_4 \approx -0.029 + 0.011i$), which do not locate its zero; an extended-precision computation is in progress.

The critical exponent. In the inner problem two growth rates are exact, from symmetries: $\lambda = 0$ (a shift of θ) and $\lambda = 1$ (a shift of the time of the singularity, the $D = \infty$ limit of the gauge mode); no symmetry gives $\lambda = 2$. Free fits of the data give $\gamma_\infty = 0.4996$–0.4998; why $\lambda_\infty = 2$ to 10^{-3} is open.

G. Relation to the next-to-leading-order condition of Ref. [11]

Ecker, Ecker and Grumiller use the same gauge: their $f = \alpha/a$ is our c, with $f = 1$ at the self-similarity horizon $x = 1$, and their profile is $\beta(\tau) \approx \sqrt{2/n}\,F(\omega\tau)$. At next-to-leading order they require $f \leq 1$ near the horizon at the zeros of β, which gives their Eq. (20); in our variables, $\omega^2 F'' + 3\omega F' = 0$ at the zeros of F. Within our framework this condition does not select Δ, for the following reasons. The inequality $f \leq 1$ holds for every regular solution, whatever the period: from $xb_x = b(p^2+m^2) - (n-1)b(b-1)$ with $b(0) = 1$, b can never drop below 1, so $xc_x = (n-1)c(b-1) \geq 0$ and $c \leq c(1) = 1$. In the formal family of Proposition 1, near a zero of F,

$$b - 1 = 2\varepsilon^2 x^2 F(\theta - \delta\theta)^2 + 2\varepsilon^4\omega^2 x^4 F'^2 + \ldots, \qquad \delta\theta(x) = -\varepsilon\omega x^2\,\frac{3F' + \omega F''}{2F'}, \tag{11}$$

so the next-to-leading-order term is the cross term of a square whose square term arrives one order later; the discriminant $16\omega^2 x^6 F'^4$ is positive for every F and ω. The double-double solutions are consistent with this: they violate Eq. (20) [$(\omega^2 F'' + 3\omega F')/(3\omega F') = 0.98$–$1.06$ at $D = 12$–60, since F is nearly antisymmetric about its zeros], the zero line of $p - m$ tilts as $\delta\theta(x)$ predicts (measured over predicted at $x = 1$: 0.69 to 0.88 for $D = 20$–60), and the maximum of c at the centre approaches $1 - (\varepsilon^3/2)\omega^2 F'^2$, their Eq. (22) in this scaling (ratio 0.36 to 0.86).

VI. DISCUSSION

The numerics give $\Delta(D)$ to high precision up to $D = 61$, and they favour a finite limit $\Delta_\infty = 1.64389375$ strongly, under the assumption of analyticity in $1/D$ that they support but cannot prove. The theory explains why that number cannot come from the $1/D$ expansion: the expansion is unobstructed for every period, and the selection acts through

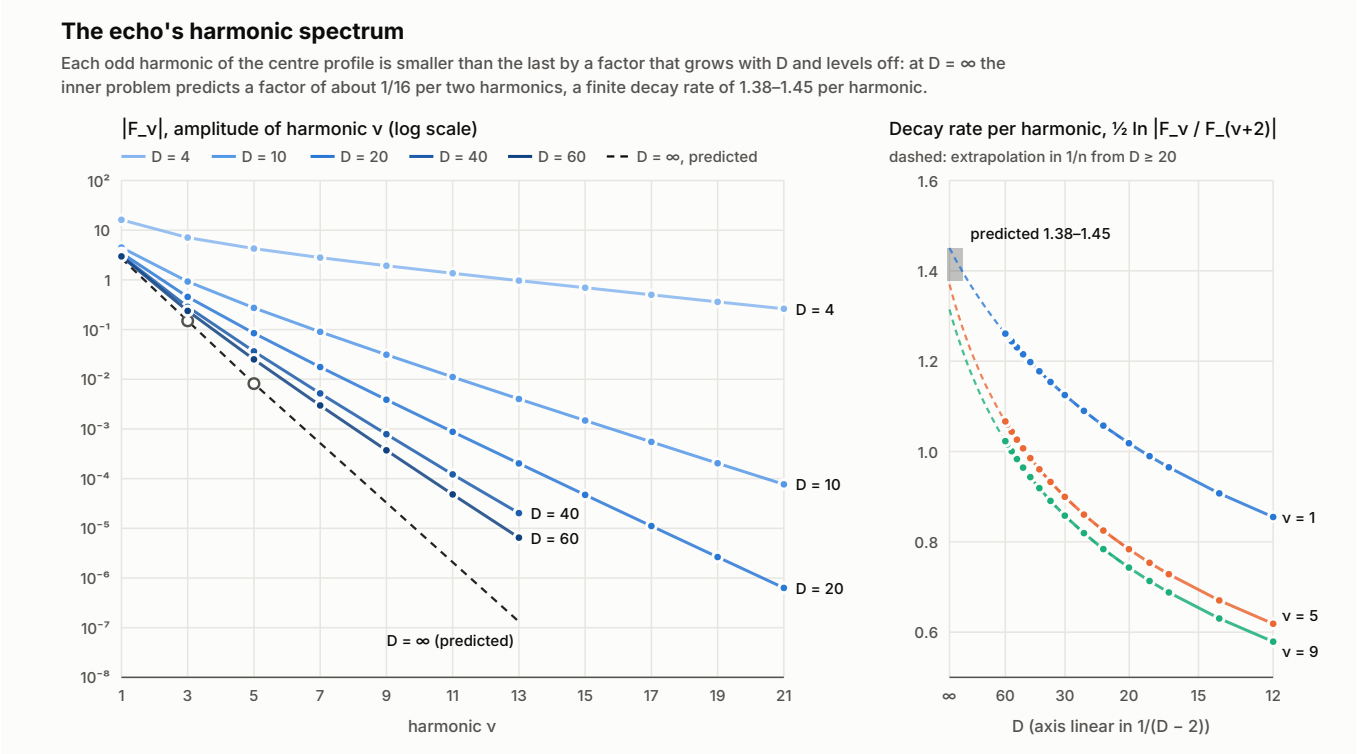


FIG. 4. Left: amplitude $|F_\nu|$ of each odd harmonic of the echo profile for $D = 4$–60 and the $D = \infty$ prediction of the inner problem (dashed; beyond $\nu = 5$ the leading-order ratio, conjectured). Right: the decay rate per harmonic, $\frac{1}{2}\ln|F_\nu/F_{\nu+2}|$, against $1/(D-2)$, with extrapolations from $D \geq 20$ and the predicted $D = \infty$ band.

an exponentially small, Stokes-type effect whose linear version is solved exactly. The inner problem at $x \sim \sqrt{n}$ is n-free; its higher-harmonic conditions already reproduce the shape of the echo at $D = \infty$. Open are: the harmonic-1 condition of the inner problem, which would give Δ_∞ (and possibly a closed form) from first principles; a proof of the working hypothesis; the value $\lambda_\infty = 2$; the $D = 5$ discrepancy; and the behaviour just below $D = 4$.

VII. HOW THIS WORK WAS DONE

N. M. chose the problem, set the direction at each step and ran the long computations on his own machine. The code, the computations, the derivations, the proofs and the text, including this paper, were produced by Claude, Anthropic's AI model, working as a set of coordinated agents. N. M. has not checked the mathematics himself, and no human expert has reviewed the work. Checks made: two independent solvers agree where they overlap; two machines with separate builds agree to 2–4×10^{-16}; every fit table reproduces from the data with the stated command; Propositions 1–3 and the inner equations were re-derived by separate agents (with sympy where possible); the inner Stokes multiplier was computed by two implementations that share no code; values at $D = 4$ and the maximum of Δ agree with the literature; and an independent agent refereed the write-up. Corrections are welcome.

Data and code. All data, code and derivation scripts are at https://github.com/nmoinvaz/dss-residual-kernel, with a reproduction script (`reproduce.sh`) and the double-double points in `hp/data/delta_dd.csv`.

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